OPEN CELL SPRAY FOAM

Dynamo 500

Light-Density Open Cell Spray Polyurethane Foam

Manufactured by **Dynamo Polyurethane Systems LLC** · 2113 Harwood Rd., Suite 309 #900, Bedford, Texas 76021 · Country of origin: **United States**

Dynamo 500 is a two-component, light-density open cell spray polyurethane foam. Open cell foam expands aggressively to fill every gap, crack and irregular cavity, forming a continuous air seal across the surface it is sprayed on. It is vapor-permeable by design, which allows an assembly to dry, and it is the material we use for the bulk of attic and cathedral ceiling depth.

R-4.1	>97%	0.5 pcf	Class I (A)
per inch	open cell content	nominal core density	flame spread rating
ASTM C518	ASTM D2856	ASTM D1622	ASTM E84

WHY WE SPECIFY IT

1. **Highest open cell R-value in its class**
   At R-4.1 per inch, Dynamo 500 outperforms typical open cell foams, which run R-3.5 to R-3.8. Over a full attic that difference adds up to real R-value at no additional depth.
2. **Seals what fiberglass cannot**
   Air permeance of 0.0013 cfm/ft^2 at 3.5 inches. The foam expands into every rafter bay, around every wire and pipe, and bonds to the sheathing. There are no gaps, seams or compressed spots.
3. **Lets the assembly dry**
   At roughly 9 perms over 2 inches, open cell foam is vapor-permeable. If moisture ever does reach the roof deck, the assembly can dry inward rather than trapping it — and a leak shows itself instead of hiding.
4. **Zero ozone-depleting blowing agents**
   Dynamo 500 is water-blown with zero ozone depletion potential and is classified as a low-VOC product.

HEALTH & INDOOR AIR QUALITY

- **Low-VOC formulation.** Dynamo 500 is classified as a low-VOC product, with worker re-entry at 1 hour and job site re-occupancy at 2 hours under ASTM D8445 at applicable ventilation rates.
- **No mold or fungal growth.** Tested to ASTM C1338 with no growth.
- **The smell is temporary and it is not toxic.** Open cell foam has a mild odor for the first 24–48 hours as it finishes curing. This is the normal scent of cured foam, not an ongoing chemical release. It dissipates completely.
- **Our standard practice with children in the home.** We recommend families with young children, elderly members, or anyone chemically sensitive plan to be out of the house during spraying and for a full 24–48 hours afterward, regardless of the manufacturer's shorter published figure.

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TECHNICAL PROPERTIES

PROPERTY	VALUE	TEST METHOD
R-Value at 1 inch	**4.1**	ASTM C518
Open-cell content	**> 97%**	ASTM D2856
Core density	**0.5 lb/ft³ nominal**	ASTM D1622
Air permeance at 3.5 in.	**0.0013 cfm/ft²**	ASTM E283
Tensile strength	**3 psi**	ASTM D1623
Water vapor transmission	**9 US perms at 2 in.**	ASTM E96
Water absorption	**33% by volume**	ASTM D2842
Dimensional stability (7 days)	**-4% at 158°F**	ASTM D2126
Fungi resistance	**No growth**	ASTM C1338
Re-entry / re-occupancy	**1 hour / 2 hours**	ASTM D8445
Flame spread	**Class I (A) — ≤ 15**	ASTM E84
Smoke development	**Class I (A) — ≤ 450**	ASTM E84

CODE COMPLIANCE & CERTIFICATIONS

Code Compliance Research Report	Intertek CCRR-0491
Construction types approved	Type I, II, III, IV, V-B — nonstructural insulation material
Unvented attics	Approved without ignition barrier — UVA walls 18 in., ceiling 18 in.
Attics & crawl spaces	ICC-ES AC377 Appendix X — meets requirements for use without a prescriptive ignition barrier
Vapor retarder	Vapor permeable. Consult local code for climate zone requirements.

ENVIRONMENTAL & CONTENT

Ozone-depleting blowing agents
None

VOC classification
Low-VOC product

MANUFACTURER DOCUMENTATION

Technical Data Sheet (TDS)	dynamosps.com → Products Tech Info → Dynamo
Safety Data Sheet — Part B resin	Dynamo SDS 500 ALL B
Safety Data Sheet — Part A / MDI ISO	Dynamo SDS ISO A
Code Evaluation Report	Intertek CCRR-0491

Full TDS and SDS for every product on your project are available on request, at no charge, same day.

COATING & ASSEMBLY NOTE

When an intumescent coating is specified over Dynamo 500, the manufacturer's approved products and rates are: as a thermal barrier, DC315 at 20 wet mils or Flame Control 60-60A at 16 wet mils; as an ignition barrier, DC315 at 4 wet mils / 3 dry mils. These are the coatings listed on the current TDS for this specific foam — a coating must be full-scale tested over the exact foam it is applied to.